Exercise 2.3

Required

Textbook p. 61

One coin toss selects the entire sample function. At the two specified times,

$$(X(1/2), X(1)) = \begin{cases} (0,-1), & \text{heads, probability } 1/2, \\ (1,2), & \text{tails, probability } 1/2. \end{cases}$$

(1) One-dimensional distribution functions

Using the right-continuous convention $F(x) = P(X \le x)$,

$$\boxed{F_X(x,1/2) = \begin{cases} 0, & x < 0, \\ 1/2, & 0 \le x < 1, \\ 1, & x \ge 1, \end{cases}} \qquad \boxed{F_X(x,1) = \begin{cases} 0, & x < -1, \\ 1/2, & -1 \le x < 2, \\ 1, & x \ge 2. \end{cases}}$$

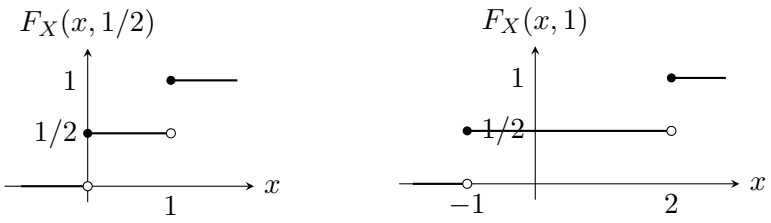
(2) Two-dimensional distribution function

An atom contributes only when both of its coordinates meet the thresholds:

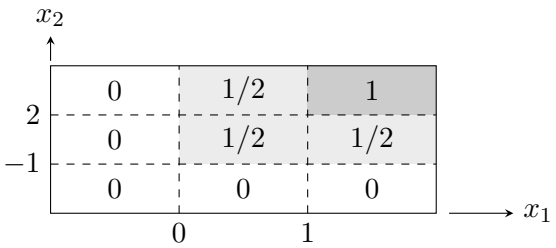
$$\boxed{F_X(x_1,x_2;1/2,1) = \tfrac{1}{2}\mathbf{1}_{\{x_1 \ge 0,\ x_2 \ge -1\}} + \tfrac{1}{2}\mathbf{1}_{\{x_1 \ge 1,\ x_2 \ge 2\}}.}$$

The two samples are not independent: they are determined by the same toss.

(3) Graphs



The joint CDF is the following step surface, shown as a level map. Each region is a horizontal plateau at the labelled height; threshold lines belong to the region on their right or above.



Outer regions extend to infinity. In the marginal graphs, filled endpoints are included.

Exercise 2.13

Required

Textbook p. 62

Let the resistance be $R > 0$ and the capacitance be C. Both branches of a parallel RC circuit have the applied voltage $X(t) = A\cos(\omega t + \Theta)$.

Total current

With the passive sign convention, the resistor and capacitor currents are

$$I_R(t) = \frac{X(t)}{R}, \qquad I_C(t) = C\frac{dX(t)}{dt}.$$

For a given sample, A and Θ are fixed with respect to time. Hence

$$\frac{dX(t)}{dt} = -A\omega\sin(\omega t + \Theta),$$

and Kirchhoff's current law gives

$$\boxed{I(t) = A\left[\frac{1}{R}\cos(\omega t + \Theta) - \omega C\sin(\omega t + \Theta)\right].}$$

Mean-square value

Write $\psi = \omega t + \Theta$. Since A and Θ are independent,

$$\mathbb{E}[I^2(t)] = \mathbb{E}[A^2]\big[R^{-2}\mathbb{E}(\cos^2\psi) + \omega^2 C^2\mathbb{E}(\sin^2\psi) - 2\omega C R^{-1}\mathbb{E}(\sin\psi\cos\psi)\big].$$

The amplitude is uniform on $[-1, 1]$, so

$$\mathbb{E}[A^2] = \frac{1}{2}\int_{-1}^{1} a^2\,da = \frac{1}{3}.$$

Averaging over one complete phase period gives

$$\mathbb{E}(\cos^2\psi) = \mathbb{E}(\sin^2\psi) = \frac{1}{2}, \qquad \mathbb{E}(\sin\psi\cos\psi) = 0.$$

For example, $\mathbb{E}(\cos^2\psi) = \frac{1}{2\pi}\int_{-\pi}^{\pi}\cos^2(\omega t + \theta)\,d\theta = 1/2$. Therefore the mean-square current is constant in time:

$$\boxed{\mathbb{E}[I^2(t)] = \frac{1}{6}\left(\frac{1}{R^2} + \omega^2 C^2\right).}$$

Exercise 2.16

Required

Textbook p. 62

Mean, covariance, and Gaussianity

Every finite collection of samples is a linear transformation of the jointly Gaussian vector (A, B); therefore $X(t)$ is a Gaussian process. Independence and zero means give $\mathbb{E}[AB] = 0$. Thus

$$\mathbb{E}[X(t)] = 0, \qquad C_X(s,t) = \sigma^2[\cos\omega s \cos\omega t + \sin\omega s \sin\omega t] = \sigma^2 \cos\omega(s-t).$$

In particular, $\mathrm{Var}[X(t)] = C_X(t,t) = \sigma^2$.

One-dimensional density

For $\sigma^2 > 0$, $X(t) \sim N(0, \sigma^2)$, so

$$\boxed{f_X(x;t) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{x^2}{2\sigma^2}\right).}$$

Two-dimensional density: nonsingular case

Let $X_1 = X(t_1)$, $X_2 = X(t_2)$ and $\rho = \cos\omega(t_1 - t_2)$. Their covariance matrix is

$$\boldsymbol{C} = \sigma^2 \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}, \quad \det \boldsymbol{C} = \sigma^4(1-\rho^2), \quad \boldsymbol{C}^{-1} = \frac{1}{\sigma^2(1-\rho^2)} \begin{bmatrix} 1 & -\rho \\ -\rho & 1 \end{bmatrix}.$$

Substitution into the bivariate normal density gives

$$\boxed{f_{X_1X_2}(x_1,x_2) = \frac{\exp\left[-\dfrac{x_1^2 - 2\rho x_1 x_2 + x_2^2}{2\sigma^2(1-\rho^2)}\right]}{2\pi\sigma^2\sqrt{1-\rho^2}}.}$$

This formula applies only when $|\rho| < 1$, equivalently $\omega(t_1 - t_2) \notin \pi\mathbb{Z}$.

Singular times

If $\omega(t_1 - t_2) = k\pi$, then $X_2 = (-1)^k X_1 = \rho X_1$ almost surely. There is no ordinary density on $\mathbb{R}^2$. A generalized representation is

$$f_{X_1X_2}(x_1,x_2) = f_X(x_1;t_1)\delta(x_2 - \rho x_1).$$

If $\sigma^2 = 0$, the entire process is zero almost surely; its one-dimensional law is $\delta(x)$.

Exercise 2.19

Required

Textbook p. 62

The mean and covariance are

$$\boldsymbol{\mu} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}, \qquad \boldsymbol{C} = \begin{bmatrix} 2 & 3 \\ 3 & 5 \end{bmatrix}.$$

Joint probability density

Compute the determinant and inverse:

$$\det \boldsymbol{C} = 2 \cdot 5 - 3^2 = 1, \qquad \boldsymbol{C}^{-1} = \begin{bmatrix} 5 & -3 \\ -3 & 2 \end{bmatrix}.$$

Since the leading principal minors are $2 > 0$ and $1 > 0$, the covariance is positive definite and the ordinary bivariate normal density exists. Let $z_1 = x - 2$, $z_2 = y - 2$. Then

$$f_{XY}(x, y) = \frac{1}{2\pi\sqrt{\det \boldsymbol{C}}} \exp\left[-\frac{1}{2} \begin{pmatrix} z_1 & z_2 \end{pmatrix} \boldsymbol{C}^{-1} \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} \right].$$

Expanding the quadratic form,

$$\boxed{f_{XY}(x, y) = \frac{1}{2\pi} \exp\left[-\frac{1}{2}\big(5(x-2)^2 - 6(x-2)(y-2) + 2(y-2)^2\big) \right],}$$

for all $x, y \in \mathbb{R}$.

Joint characteristic function

For a Gaussian vector and $\boldsymbol{q} = (u, v)^{\mathrm{T}}$,

$$\phi_{XY}(u, v) = \mathbb{E}[\mathrm{e}^{\mathrm{j}(uX+vY)}] = \exp\left(\mathrm{j}\boldsymbol{q}^{\mathrm{T}}\boldsymbol{\mu} - \frac{1}{2}\boldsymbol{q}^{\mathrm{T}}\boldsymbol{C}\boldsymbol{q} \right).$$

Here

$$\boldsymbol{q}^{\mathrm{T}}\boldsymbol{\mu} = 2u + 2v, \qquad \boldsymbol{q}^{\mathrm{T}}\boldsymbol{C}\boldsymbol{q} = 2u^2 + 6uv + 5v^2.$$

Therefore

$$\boxed{\phi_{XY}(u, v) = \exp\left[\mathrm{j}(2u + 2v) - u^2 - 3uv - \frac{5}{2}v^2 \right].}$$

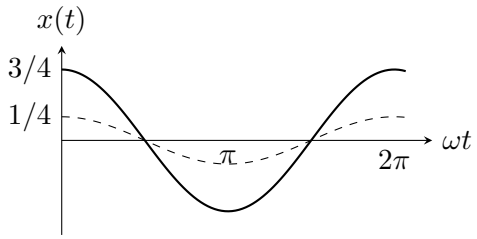
Exercise 2.1

Optional

Textbook p. 61

(1) Two sample functions

Fixing a realization $V = v$ gives $x(t) = v\cos\omega t$. For example, take $v = 1/4$ and $v = 3/4$.



(2) Densities at the four specified times

Write $c = \cos\omega t$. For $c \neq 0$, the transformation $x = cv$ has inverse $v = x/c$ and absolute Jacobian $1/|c|$. Since $f_V(v) = 1$ on $[0, 1]$,

$$f_X(x;t) = \frac{1}{|c|}\mathbf{1}_{\{\min(0,c)\le x\le\max(0,c)\}}.$$

Applying this rule gives

Time t	$c = \cos\omega t$	Density (zero outside the stated interval)
0	1	$1, \quad 0 \le x \le 1$
$\pi/(4\omega)$	$1/\sqrt{2}$	$\sqrt{2}, \quad 0 \le x \le 1/\sqrt{2}$
$3\pi/(4\omega)$	$-1/\sqrt{2}$	$\sqrt{2}, \quad -1/\sqrt{2} \le x \le 0$
π/ω	-1	$1, \quad -1 \le x \le 0$

The negative values of c reverse the support interval; the density coefficient remains positive.

(3) Density at $t_0 = \pi/(2\omega)$

Here $c = 0$, so $X(t_0) = 0$ for every realization of V. The law is a point mass:

$$\boxed{f_X(x;t_0) = \delta(x), \qquad P\{X(t_0) = 0\} = 1.}$$

This is a generalized density; the $1/|c|$ formula cannot be used at $c = 0$. The listed nonnegative times implicitly require $\omega > 0$.

Exercise 2.4

Optional

Textbook p. 61

A Bernoulli random sequence has independent bits, with $P(B(n) = 1) = 0.8$ and $P(B(n) = 0) = 0.2$.

(1) Probability of the word 1011

Independence allows the four probabilities to be multiplied:

$$P\{(B(n), B(n+1), B(n+2), B(n+3)) = (1, 0, 1, 1)\} = 0.8 \cdot 0.2 \cdot 0.8 \cdot 0.8 = \boxed{0.1024}.$$

(2) Mean word

Interpret the four-bit word as the random vector

$$\boldsymbol{B} = (B(n), B(n+1), B(n+2), B(n+3)).$$

Each component has mean $\mathbb{E}[B(n)] = 0(0.2) + 1(0.8) = 0.8$. Therefore

$$\boxed{\mathbb{E}[\boldsymbol{B}] = (0.8, 0.8, 0.8, 0.8).}$$

This is the ensemble mean vector, not an actual binary realization. If instead a word is interpreted as an unsigned binary integer, its expected value is $\mathbb{E}[8B(n) + 4B(n+1) + 2B(n+2) + B(n+3)] = 15(0.8) = 12$.

(3) Most probable word

A particular word containing k ones has probability

$$0.8^k 0.2^{4-k}.$$

Replacing a zero by a one multiplies this probability by $0.8/0.2 = 4 > 1$. Thus the maximum occurs when all four bits are one:

$$\boxed{\text{Most probable word: } 1111, \qquad P(1111) = 0.8^4 = 0.4096.}$$

There is no binomial coefficient for a specified word. The factor $\binom{4}{k}$ appears only when summing over all words containing k ones.

Exercise 2.7

Optional

Textbook p. 61–62

The process is $X(t) = Vt + X_0$, where $V \sim N(1,1)$ and $X_0 \sim N(0,2)$ are independent. In $N(\mu, \sigma^2)$, the second parameter is the variance.

Mean

At any fixed time t, linearity of expectation gives

$$\boxed{m_X(t) = \mathbb{E}[X(t)] = t\mathbb{E}[V] + \mathbb{E}[X_0] = t.}$$

Variance

Independence implies $\mathrm{Cov}(V, X_0) = 0$, so

$$\begin{aligned}\mathrm{Var}[X(t)] &= t^2 \,\mathrm{Var}(V) + \mathrm{Var}(X_0) + 2t \,\mathrm{Cov}(V, X_0) \\ &= t^2 + 2.\end{aligned}$$

Thus

$$\boxed{\sigma_X^2(t) = t^2 + 2.}$$

One-dimensional density

The linear combination of the independent Gaussian variables is Gaussian. One can verify this directly using the characteristic function:

$$\begin{aligned}\phi_{X(t)}(u) &= \mathbb{E}[\mathrm{e}^{\mathrm{j}u(tV+X_0)}] = \phi_V(tu)\phi_{X_0}(u) \\ &= \exp\left(\mathrm{j}tu - \frac{1}{2}t^2u^2\right)\exp(-u^2) \\ &= \exp\left(\mathrm{j}tu - \frac{1}{2}(t^2+2)u^2\right).\end{aligned}$$

This is the characteristic function of $N(t, t^2+2)$. Therefore

$$\boxed{f_X(x;t) = \frac{1}{\sqrt{2\pi(t^2+2)}}\exp\left[-\frac{(x-t)^2}{2(t^2+2)}\right], \qquad x \in \mathbb{R}.}$$

As a check, at $t = 0$ this reduces to the density of $X_0 \sim N(0,2)$.

Exercise 2.8

Optional

Textbook p. 62

For each n, $P(I_n = -1) = P(I_n = 1) = 1/2$, and the Bernoulli sequence has independent samples.

One-dimensional probability density

The distribution is discrete, so its density is expressed using Dirac impulses:

$$\boxed{f_{I_n}(x) = \frac{1}{2}\delta(x+1) + \frac{1}{2}\delta(x-1).}$$

Each coefficient is a probability mass. There is no ordinary continuous density.

Mean

$$\boxed{m_I(n) = \mathbb{E}[I_n] = (-1)\frac{1}{2} + (1)\frac{1}{2} = 0.}$$

Since $I_n^2 = 1$ for every outcome,

$$\mathbb{E}[I_n^2] = 1, \qquad \mathrm{Var}(I_n) = 1.$$

Covariance function

By definition,

$$C_I(n,m) = \mathbb{E}[(I_n - \mathbb{E}I_n)(I_m - \mathbb{E}I_m)] = \mathbb{E}[I_n I_m].$$

When $n = m$, this equals $\mathbb{E}[I_n^2] = 1$. When $n \neq m$, independence gives

$$\mathbb{E}[I_n I_m] = \mathbb{E}[I_n]\mathbb{E}[I_m] = 0.$$

Consequently,

$$\boxed{C_I(n,m) = \begin{cases} 1, & n = m, \\ 0, & n \neq m, \end{cases} = \delta_{nm}.}$$

Here δ_{nm} is the Kronecker delta for discrete indices. It is distinct from the Dirac delta $\delta(x)$ used in the density. Since the mean is zero, $R_I(n,m) = C_I(n,m)$ as well.

Exercise 2.11

Optional

Textbook p. 62

The stated assumption needs care

The textbook states *pairwise* independence of A, B, Θ. This does not imply independence of AB and Θ. Assume the moments below exist. Pairwise independence does give $\mathbb{E}[X(t)] = \mathbb{E}[Y(t)] = 0$, because each amplitude is independent of the uniform phase.

(1) Cross-correlation under the stated assumptions

Set $\alpha = \omega t_1$, $\beta = \omega t_2$. The identity $\sin a \cos b = \frac{1}{2}[\sin(a-b) + \sin(a+b)]$ yields

$$\boxed{R_{XY}(t_1, t_2) = \tfrac{1}{2}\mathbb{E}[A]\mathbb{E}[B]\sin(\alpha - \beta) + \tfrac{1}{2}\mathbb{E}[AB\sin(\alpha + \beta + 2\Theta)].}$$

The last expectation is not determined by pairwise independence. Thus the stated information does not specify a unique cross-correlation.

For a concrete counterexample, let Z be an independent equiprobable sign, let $h(\Theta) = \operatorname{sgn}(\sin 2\Theta)$, and set $A = Z$, $B = Zh(\Theta)$. Both A and B are independent of Θ; A and B are also independent since h is an equiprobable sign. Yet $AB = h(\Theta)$, and

$$R_{XY}(0,0) = \mathbb{E}[h(\Theta)\sin\Theta\cos\Theta] = \tfrac{1}{2}\mathbb{E}|\sin 2\Theta| = \frac{1}{\pi} \neq 0.$$

Values of h at its finitely many zeros do not affect these probabilities.

Common intended answer under the stronger assumption

If A, B, Θ are *mutually* independent (or (A, B) is independent of Θ, in addition to $A \perp B$), the phase average of the last term is zero. Then

$$\boxed{R_{XY}(t_1, t_2) = \tfrac{1}{2}\mathbb{E}[A]\mathbb{E}[B]\sin\omega(t_1 - t_2).}$$

This extra assumption is needed for this familiar formula; it is not silently inferred from the question.

(2) Orthogonality, uncorrelatedness, and independence

Since both means are zero, $C_{XY} = R_{XY}$. Hence orthogonality and uncorrelatedness are equivalent here, but cannot be decided from the stated information alone. Under the stronger assumption and $\omega \neq 0$, they hold for all pairs of times exactly when $\mathbb{E}[A]\mathbb{E}[B] = 0$. Equal-time orthogonality alone, $R_{XY}(t,t) = 0$, is weaker than orthogonality for every pair of times.

Statistical independence does not follow even when $R_{XY} \equiv 0$. For example, take mutually independent equiprobable signs A, B and an independent uniform phase. Then $\mathbb{E}A = \mathbb{E}B = 0$, but $X(t)^2 + Y(t)^2 = 1$. More explicitly, each of $|X(t)| > \sqrt{3}/2$ and $|Y(t)| > \sqrt{3}/2$ has probability $1/3$, while their intersection is impossible. Thus even the equal-time variables are dependent. Independence is possible in degenerate cases such as $A \equiv 0$; the given assumptions do not settle it.

Exercise 2.18

Optional

Textbook p. 62

Mean vector and covariance matrix

Write $X_0 = X(0)$, $X_1 = X(1/2)$, $X_2 = X(1)$, and set $c = \cos(1/2)$, $s = \sin(1/2)$. Then

$$\boldsymbol{\mu} = \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix}, \qquad \boldsymbol{C} = 8 \begin{bmatrix} 1 & c & \cos 1 \\ c & 1 & c \\ \cos 1 & c & 1 \end{bmatrix}.$$

Why the ordinary three-dimensional density does not exist

Let $\boldsymbol{a} = (1, c, \cos 1)^{\mathrm{T}}$ and $\boldsymbol{b} = (0, s, \sin 1)^{\mathrm{T}}$. The cosine difference identity gives

$$\boldsymbol{C} = 8(\boldsymbol{a}\boldsymbol{a}^{\mathrm{T}} + \boldsymbol{b}\boldsymbol{b}^{\mathrm{T}}).$$

Thus $\operatorname{rank}(\boldsymbol{C}) \leq 2$. Its upper-left 2×2 determinant is $64s^2 > 0$, so its rank is exactly 2, and $\det \boldsymbol{C} = 0$. The vector $\boldsymbol{q} = (1, -2c, 1)^{\mathrm{T}}$ satisfies $\boldsymbol{C}\boldsymbol{q} = 0$. Therefore

$$\operatorname{Var}\big[(X_0 - 2) - 2c(X_1 - 2) + (X_2 - 2)\big] = 0,$$

so the centered expression is zero almost surely. All probability lies on a plane:

$$\boxed{(X_0 - 2) - 2c(X_1 - 2) + (X_2 - 2) = 0.}$$

Consequently, an ordinary density with respect to three-dimensional volume cannot be written. The nonsingular Gaussian PDF formula is inapplicable.

A valid generalized density

Put $z_i = x_i - 2$. The pair (X_0, X_1) is nonsingular Gaussian with covariance $8 \begin{bmatrix} 1 & c \\ c & 1 \end{bmatrix}$. Combining its density with the plane constraint gives

$$\boxed{f(x_0, x_1, x_2) = \frac{1}{16\pi|s|} \exp\left[-\frac{z_0^2 - 2cz_0z_1 + z_1^2}{16s^2}\right] \delta(z_2 - 2cz_1 + z_0).}$$

Integration over x_2 removes the delta with unit Jacobian, leaving a normalized bivariate density. Equivalently, a description valid without a PDF is

$$\boxed{\phi(\boldsymbol{u}) = \exp\left(\mathrm{j}\boldsymbol{u}^{\mathrm{T}}\boldsymbol{\mu} - \tfrac{1}{2}\boldsymbol{u}^{\mathrm{T}}\boldsymbol{C}\boldsymbol{u}\right).}$$

The requested ordinary 3D PDF does not exist for the covariance given in the textbook.